Khanh An Nguyen

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EDUCATION

Monash University **Mar. 2026 - May. 2029 (expected)**
Degree: Bachelor of Engineering (Software Engineering)

PROJECTS

Personal Website: https://nguyenkhanhan.site

C++ Game Boy Emulator (DMG-01):

- Implemented all 512 SM83 CPU instructions with correct flag behavior, addressing modes, and interrupt handling
- Reproduced hardware rendering behavior for background, window, and sprite layers through development of a cycle-accurate PPU using a scanline + Pixel FIFO pipeline
- Achieved real-time emulation (~59.7 FPS) through optimized instruction dispatch and frame timing control
- Enable compatibility with ~90% of commercial Game Boy titles through implementation of MBC1, MBC2, MBC3, and MBC5 cartridge controllers

Full-Stack Chess Web Application:

- Built a real-time multiplayer chess platform using React (frontend) and Spring Boot (backend) with PostgreSQL for persistent game and user data
- Designed REST APIs and WebSocket-based game synchronization, enabling low-latency move updates and live board state consistency between players
- Implemented JWT-based authentication and authorization, securing user accounts and game actions
- Containerized the system with Docker, enabling reproducible deployment of frontend, backend, and database services

WORK EXPERIENCES

Electric Motorcycle World **Mar. 2025 - Dec. 2026**
Freelance Software Engineer

- Reduced price lookup time by 40% by engineering a GUI-based data pipeline to transform and consolidate multiple XLSX product sheets into a structured SQL database
- Developed a mobile workforce management application to track employee payroll schedules, improving payroll accuracy and reducing administrative overhead

DataAnnotation (Contract) **Mar 2025 – Nov 2025**
LLM Response Reviewer

- Evaluated and benchmarked outputs from 1-4 LLM models for each task, analyzing accuracy, coherence, and relevance while uncovering edge cases, biases, and response inconsistencies.
- Designed targeted prompt variations to stress-test model behavior across diverse scenarios, delivering structured feedback that improved output quality and failure detection.

SKILLS

- **Languages**: English, Vietnamese
- **Programming Languages:** Java, Javascript, Typescript, HTML/CSS, Python, C++
- **Frameworks:** Spring, ReactJS
- **Database**: PostgreSQL
- **Interests:** Reading Books, Solving Coding Puzzles, British TV shows, Pokemon

CERTIFICATIONS & ACHIEVEMENTS

- **IELTS Academic** - Overall Band Score: 8.0
- **ISTQB Certified Tester** - Foundation Level (CTFL)
- **BOSCH CodeRace Challenge 2025** - Top 20